Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

CHEMISTRY

Class – 6

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1. Fibre and Fabric

Solutions

Worksheet – 1

1. **Correct answer – D**
   Cotton grows best in black soil which retains moisture and requires a warm climate for proper growth.
2. **Correct answer – A**
   Fibres are broadly classified into natural fibres (cotton, wool, silk) and synthetic fibres (nylon, polyester).
3. **Correct answer – B**
   Silk is produced from the cocoon of silkworms.
4. **Correct answer – C**
   Jute needs heavy rainfall and warm climate, so it is grown during the rainy season.
5. **Correct answer – B**
   Fibres are thin strands that are spun into yarn and then woven into cloth.
6. **Correct answer – B**
   Synthetic fibres are man-made fibres produced from chemical substances in factories (e.g., nylon, polyester).
7. **Correct answer – A**
   Cotton requires black soil and a warm, dry climate. Assam has heavy rainfall, which is unsuitable for cotton cultivation.
8. **Correct answer – A**
   Fibres obtained from plants (cotton, jute) and animals (wool, silk) are called natural fibres.
9. **Correct answer – C**
   Flax is a natural plant fibre, whereas nylon, polyester, and orlon are synthetic fibres.
10. **Correct answer – B**
    Silk is obtained from the cocoon of silkworms, which are animals.

Worksheet – 2

1. **Correct answer – C**
   Fabric is made by weaving or knitting yarn, which consists of long, twisted threads obtained from fibres.
2. **Correct answer – C**
   Sweaters are made by knitting loops of yarn together.
3. **Correct answer – C**
   Knitting involves interlocking loops of a single yarn using needles or machines.
4. **Correct answer – A**
   Spinning is the process by which fibres are twisted together to form yarn.
5. **Correct answer – C**
   Linen fibres are obtained from the stem of the flax plant.
6. **Correct answer – C**
   Gadwal is a well-known handloom centre in Telangana.
7. **Correct answer – D**
   A power loom operates using electric power, unlike a handloom which is manually operated.
8. **Correct answer – A**
   Cotton is a plant fibre and burns with a smell similar to burning paper.
9. **Correct answer – A**
   Yarn is a spun thread made from fibres and is used for weaving or knitting fabrics.
10. **Correct answer – B**
    In ancient times, people used natural materials like bark, leaves, and animal skins for
    clothing before fibres such as cotton and silk were discovered or processed.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**
   Fabrics are mainly used to make clothes for protection, comfort, and appearance.
2. **Correct answer – D**
   Flax is a plant fibre obtained from the stem of the flax plant, whereas wool and silk are animal fibres and nylon is synthetic.
3. **Correct answer – D**
   Silk is obtained from the cocoon spun by silkworms.
4. **Correct answer – C**
   Acrylic is a synthetic fibre that is often used as a substitute for wool in winter clothing.
5. **Correct answer – C**
   Retting is the process of soaking jute stems in water to separate the fibres.
6. **Correct answer – B**
   Silk yarn is obtained from the cocoon formed by silkworms.
7. **Correct answer – B**
   Nylon's high strength and elasticity make it suitable for durable items.
8. **Correct answer –B**
   Cotton absorbs sweat and allows air circulation, which helps keep the body cool in hot weather.
9. **Correct answer – B**
   Weaving involves interlacing two sets of yarn at right angles to create fabric.
10. **Correct answer – A**
    Fabric is made by weaving yarns together.
11. **Correct answer – B**
    Palm leaves can be used to make mats.
12. **Correct answer – D**
    Venkatagiri is the famous handloom centre in Andhra Pradesh.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

13. **Correct answer – A, C, D**
    Nylon is synthetic.
14. **Correct answer – A, B, C**
    Silk is an animal fibre.
15. **Correct answer – A, B, C**
    Wool is a natural animal fibre.
16. **Correct answer – A and B**
    Weaving and knitting make fabric; ginning separates cotton fibres from seeds. Spinning makes yarn from fibres.
17. **Correct answer – A, B, C**
    Scissors are used only for cutting.

ASSERTION AND REASNING TYPE

18. **Correct answer – A**
    Polyester dries quickly and is easy to maintain because it is a synthetic fibre that does not absorb sweat well. Hence, both the assertion and reason are true, and the reason correctly explains the Assertion.
19. **Correct answer – C**
    Knitting uses a single yarn, so the Assertion is true. The reason is false because interlacing two sets of yarns happens in weaving, not knitting.
20. **Correct answer – B**
    The Assertion is true because natural fibres come from natural sources like plants and animals. The Reason is also true since nylon is a synthetic fibre, but it does not explain why natural fibres are obtained from natural sources.
21. **Correct answer – D**
    Cotton fibres are separated from seeds by ginning, not spinning. Therefore, the assertion is false, but the reason is true.
22. **Correct answer – A**
    Jute is called the golden fibre because of its golden colour and its high economic value due to widespread use in bags, carpets, ropes, and other products. Hence, both the assertion and reason are true, and the reason correctly explains the assertion.

MATRIX MATCH TYPE

23. Correct answer

a-Q, b-R, c-S, d-P

Wool is obtained from the hair (fleece) of sheep. Linen fibres are obtained from the stem of the flax plant. Silk is produced from the cocoon spun by silkworms. Cotton fibres are obtained from the cotton plant.

24. Correct answer

a-R, b-P, c-S, d-Q

Retting loosens fibres from plant stems by soaking them in water. Weaving uses two sets of yarns to form fabric. Knitting forms fabric by interlocking loops of one yarn. Spinning converts fibres into yarn by twisting them together.

COMPREHENSHION TYPE

Passage - I

25. Correct answer – C

Power-operated looms run on electricity to weave yarn into fabric.

26. Correct answer – B

Power-operated looms are mainly used in factories for large-scale production.

27. Correct answer – C

Power-operated looms are mainly used in factories for large-scale production.

Passage – II

28. Correct answer – C

Cotton absorbs water easily because it is a natural plant fibre.

29. Correct answer – B

Drying time is a characteristic of fabrics.

30. Correct answer – C

After burning, natural fibres leave ash, while synthetic fibres may form hard beads.

Previous Year Questions

1. Correct answer – B

Y lies in the overlap of plant fibre and obtained from stem. Linen is a plant fibre obtained from the stem of the flax plant, so it correctly fits position Y.

2. Correct answer – D

Cotton burns with a smell similar to burning paper, just as wool burns with a smell like burning hair.

3. Correct answer – C

In Group III: Polyester is synthetic. Cotton and fur are natural. This is mixed group. Hence this grouping is not correct.

4. Correct answer – A

Jute fibre is not obtained from the leaf; it is obtained from the stem of the jute plant. Therefore, the statement saying "Leaf should not be replaced" is incorrect.

5. Correct answer – C

Wool is obtained from sheep and also from animals like goat, yak, camel, etc. Rearing of silkworm for silk production is called sericulture. Corchorus olitorius is the jute plant; cotton comes from the cotton plant. The process of soaking jute stalks in water to let these rot for obtaining jute fibre is called retting.

6. Correct answer – D

weaving and knitting can be done by hand or by machines. weaving uses two sets of yarns, while knitting uses one yarn. weaving and knitting are not limited to specific fibres like silk or wool only.

7. Correct answer – B

Coir is obtained from the fruit (coconut). Hemp is a bast fibre obtained from the stem of the plant. Sisal is obtained from the leaves of the sisal plant.

8. Correct answer – C

The picture shows cotton being combed to remove seeds from the cotton fibres. This process is part of ginning, where seeds are separated before the fibres are spun into yarn.

9. Correct answer – C

X shows a Takli (hand spindle), used for spinning fibres into yarn. X shows a Charkha (spinning wheel), also used for spinning yarn. Hence, X = Takli and Y = Charkha.

10. Correct answer – B

Fabrics are made of yarns, and yarns are made up of fibres. Hence, L = Fabrics and M = Fibre.

2. Nature of Matter

Solutions

Worksheet – 1

1. **Correct Answer: C) Matter and radiation**
   Explanation: The universe consists of everything that has mass (matter) and the energy that travels through space (radiation, like light).
2. **Correct Answer: C) Sunlight**
   Explanation: Sunlight is a form of electromagnetic radiation. Wood, water, and stone are all forms of matter because they have mass and take up space.
3. **Correct Answer: B) Growth and reproduction**
   Explanation: These are two "signs of life." Living things also perform respiration, have a metabolism, and respond to their environment.
4. **Correct Answer: C) Toy**
   Explanation: Mountains, rivers, and air exist naturally. A toy is manufactured by humans using various materials (matter).
5. **Correct Answer: C) Molecules of elements**
   Explanation: When atoms of the same type join (like two Oxygen atoms), they form a molecule of an element. If they were different atoms, they would form a compound.
6. **Correct Answer: C) O_2**
   Explanation: O_2 consists only of Oxygen atoms. H_2O, CO_2, and NH_3 are all compounds because they contain two or more different elements.
7. **Correct Answer: B) Atoms in one molecule**
   Explanation: Atomicity describes how many atoms are bonded together in a single molecule of that substance.
8. **Correct Answer: B) 2**
   Explanation: Valency is the combining capacity of an atom. Oxygen needs 2 electrons to complete its outer shell, so its valency is 2.
9. **Correct Answer: C) 4**
   Explanation: The formula P_4 shows that there are four phosphorus atoms bonded together in one molecule.
10. **Correct Answer: B) Molecules and atoms**
    Explanation: Atoms are the basic building blocks of all matter. When atoms bond together, they form molecules.

Worksheet-2

1. **Correct Answer: C) Gases**
   Explanation: Gases have the highest compressibility because their particles are very far apart. Applying pressure easily pushes them closer together.
2. **Correct Answer: C) Gases**
   Explanation: "Intermolecular space" refers to the distance between particles. In gases, these distances are vast compared to the size of the particles themselves.
3. **Correct Answer: C) Very Loosely**
   Explanation: Molecules in gases are packed very loosely and move randomly. Solids are packed "Very closely," and liquids are "Loosely."
4. **Correct Answer: C) Hold particles together**
   Explanation: Cohesive forces are the attractive forces between like molecules (e.g., water molecule to water molecule) that keep a substance from falling apart.

5. **Correct Answer: B) Always in motion**
Explanation: Brownian motion is the zig-zag, random movement of microscopic particles suspended in a fluid, proving that atoms/molecules are constantly moving.
6. **Correct Answer: C) Evenly mixed**
Explanation: Diffusion is the movement of particles from high to low concentration. It continues until equilibrium is reached (uniform distribution).
7. **Correct Answer: B) Subscripts**
Explanation: Subscripts are the small numbers written at the bottom right of a symbol (like the 2 in H_2O) to show how many atoms of that element are present.
8. **Correct Answer: B) 2**
Explanation: In CO_2, the subscript 2 follows the 'O', indicating there are two Oxygen atoms for every one Carbon atom.
9. **Correct Answer: B) Oxygen**
Explanation: Iron, Magnesium, and Sodium are metals (conductors, shiny). Oxygen is a gas at room temperature and is a non-metal.
10. **Correct Answer: C) K_2SO_3**
Explanation: Potassium has a valency of +1 and Sulphite (SO_3) has a valency of -2. Using the "criss-cross" method, the 2 goes to K, resulting in K_2SO_3.

Worksheet-3

1. **Correct Answer: C) Interconversion of matter**

Explanation: This is the general term for the process where matter changes from one state to another (and back) by changing physical conditions like temperature or pressure.
2. **Correct Answer: C) Liquid water**
Explanation: When ice (solid) is heated, it gains energy and its particles begin to move more freely, changing it into liquid water
3. **Correct Answer: C) Melting**
Explanation: Melting is the phase change from solid to liquid. This occurs at a specific temperature called the melting point.

4. **Correct Answer: C) Vaporisation**
Explanation: Vaporisation (or boiling) is the process where a liquid turns into a gas by absorbing heat energy.
5. **Correct Answer: B) 100°C**
Explanation: At sea level, pure water boils and turns into steam at exactly 100°C..
6. **Correct Answer: C) Liquid**
Explanation: Cooling removes energy from gas particles. They slow down and move closer together, turning back into a liquid.
7. **Correct Answer: C) Condensation**
Explanation: Condensation is the opposite of vaporisation. It is the process where a gas or vapor changes into a liquid.
8. **Correct Answer: C) 0°C**
Explanation: Water freezes into ice at 0°C. Interestingly, this is the same temperature at which ice begins to melt.
9. **Correct Answer: C) Sublimation**
Explanation: This is a special process where a solid skips the liquid phase entirely and turns directly into a gas (e.g., dry ice or naphthalene balls).
10. **Correct Answer: B) Solidification**
Explanation: Freezing involves a liquid turning into a "solid" mass, which is why it is scientifically referred to as solidification.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: C) Mass**

Explanation: Mass is the actual measurement of the amount of matter in an object. Volume is the space it takes up, and Density is mass per unit volume.

2. **Correct Answer: D) Sun**
   Explanation: The Sun emits electromagnetic radiation (light and heat). Air, water, and books are all matter because they have mass and volume.
3. **Correct Answer: B) Forces in gases are minimum**
   Explanation: In gases, particles are so far apart that the attraction between them is almost zero. Conversely, solids have maximum attraction and minimum space.
4. **Correct Answer: B) Feeling of cold**
   Explanation: Matter must have mass and occupy space. A "feeling" is a sensory perception/energy state, not a physical substance.
5. **Correct Answer: B) Liquids can be compressed.**
   Explanation: This statement is backwards. Gases are easily compressed, while liquids are almost incompressible because their particles are already close together.
6. **Correct Answer: D) Gases in Atmosphere**
   Explanation: LPG, Oxygen cylinders, and CNG all rely on compressing huge volumes of gas into small tanks. The general presence of the atmosphere is not a "human application" of compression.
7. **Correct Answer: D) Both 'B' and 'C'**
   Explanation: Perfumes work because the gas particles expand to fill the room and diffuse (mix) with the air to reach your nose.
8. **Correct Answer: B) Iodine**
   Explanation: At room temperature, Iodine is a solid, meaning its intermolecular forces are strong enough to hold particles in a fixed structure. The others are gases (Cl, F, HCl).
9. **Correct Answer: A) Diffusibility**
   Explanation: Gases mix completely and uniformly (homogeneously) because their particles move constantly and freely to occupy all available space (diffusion).
10. **Correct Answer: D) Carbon, Potassium**
    Explanation: Both Carbon and Potassium are solids at room temperature. Solids have much stronger intermolecular forces than liquids (like Mercury/Bromine) or gases.
11. **Correct Answer: B) Close packing**
    Explanation: The universe consists of everything that has mass (matter) and the energy that travels through space (radiation, like light).
12. **Correct Answer: C) Iron**
    Explanation: Fe is the chemical symbol for Ferrum (Iron). $FeSO_3$ is specifically Ferrous Sulphite.
13. **Correct Answer: D) 4**
    Explanation: Silicon is in Group 14 of the periodic table; it has 4 valence electrons and a valency of 4.
14. **Correct Answer: D) Metal**
    Explanation: Sodium (Na) is a highly reactive Alkali Metal.
15. **Correct Answer: C) 1**
    Explanation: In CH_4 (Methane), there is no subscript after the C, which mathematically implies 1 Carbon atom.
16. **Correct Answer: C) 3**
    Explanation: Atomicity is the count of atoms in a molecule. In O_3, there are 3 oxygen atoms.
17. **Correct Answer: C) Molecules of elements**
    Explanation: As seen in previous sets, atoms of the same element (like O + O) form an elemental molecule.
18. **Correct Answer: C) Mountain**
    Explanation: Mountains occur in nature but are not "alive." Trees, bacteria, and humans are all living biological entities.
19. **Correct Answer: C) 100°C**
    Explanation: For a pure substance, the Boiling Point and Condensation Point are the same temperature. Water turns to steam at 100°C, and steam turns to water at 100°C.
20. **Correct Answer: B) Sublimate**
    Explanation: In the process of sublimation, the solid that turns into gas is the "sublime" substance, and the solid collected after cooling the vapor is called the sublimate.

21. **Correct Answer: C) Run hot water over the metal lid only.**
    Explanation: Hot water makes the metal lid expand faster than the glass, loosening the grip.
22. **Correct Answer: B) A chemical change because new substances are formed.**
    Explanation: When limestone (CaCO3) is heated, it breaks down into new substances: quicklime (CaO) and carbon dioxide (CO2).

- New substances are formed → chemical change
- It's not just a physical change because the original substance doesn't stay the same.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

23. **Correct Answer: All of the above (A, B, C, D)**
    Explanation: This is likely a multiple-choice question where all options are correct. Gases are compressible (can be squashed), expansible (fill any container), exert pressure equally in all directions, and have low density due to the large space between particles.
24. **Correct Answer: All of the above (A, B, C, D)**
    Explanation: By definition, matter is anything that has mass and volume (occupies space). Additionally, the Kinetic Theory confirms all matter is made up of tiny particles (atoms/molecules).
25. **Correct Answer: B) 273K**
    Explanation: On the Celsius scale, water freezes at $0^{\circ}C$. To convert to Kelvin, we use K = $^{\circ}C$ + 273. Therefore, 0 + 273 = 273K. (Note: 373K is the boiling point).
26. **Correct Answer: B) Iron rod & C) Wooden box**
    Explanation: Both have a definite shape and volume. Sunlight is radiation (energy), and Bromine is one of the only two elements that is liquid at room temperature.
27. **Correct Answer: A) Temperature & B) Pressure**
    Explanation: To change a state (e.g., gas to liquid), you can either lower the temperature (cool it) or increase the pressure (compress it). Changing color does not change the state of matter.
28. **Correct Answer: A) Solid to liquid & B) Liquid to gas**
    Explanation: This is a two-step process. First, ice reaches $0^{\circ}C$ and melts **(**solid to liquid). As you continue heating, the water reaches $100^{\circ}C$ and boils (liquid to gas).

ASSERTION AND REASNING TYPE

29. **Correct Answer: D) Assertion is false, but Reason is true.**
    Explanation: The Reason is correct: Condensation is indeed gas turning into liquid. However, the Assertion is false because condensation happens when heat is released (lost), not absorbed. Particles must lose energy to slow down and clump together into a liquid.
30. **Correct Answer: A) Both A and R are true, and R is the correct explanation.**
    Explanation: The Assertion is true because the state (Solid, Liquid, Gas) is determined by how molecules are arranged. The Reason correctly explains this by describing the specific packing (tight in solids, loose in gases) that creates those states.
31. **Correct Answer: D) Assertion is false, but Reason is true.**
    Explanation: The Assertion is false; solids have almost zero compressibility. The Reason is true: they are tightly packed. In fact, because they are so tightly packed (Reason), they cannot be compressed (making the Assertion false).

32. **Correct Answer: A) Both A and R are true, and R is the correct explanation.**

Explanation: The Assertion is true; a liter of water is always a liter (volume), but it takes the shape of its container. The Reason explains this: because particles can move past each other (flow), they don't have a fixed shape.

MATRIX MATCH TYPE

33. **Correct Answer: a-P;b-Q,c-R;d-S**

Explanation:

a) Evaporation is the change of a liquid into vapour at temperatures below its boiling point

b) Condensation is the process in which gas changes into liquid on cooling.

c) Sublimation is the direct change of solid into gas without becoming liquid (e.g., camphor).

d) Fusion means melting — the change of solid into liquid on heating.

34. **Correct Answer: a-S;b-P,c-Q;d-R**

Explanation:

a) These metals are precious, shiny, and do not corrode easily, so they are used for jewellery.

b) Graphite is a form of carbon used in pencil leads because it is soft and leaves marks on paper.

c) Salt is used to preserve food and enhance taste.

d) Vinegar is used in cooking, cleaning, and even gardening (e.g., removing weeds).

COMPREHENSHION TYPE

35. **Correct Answer: B) Freezing**

Explanation:

Freezing is the process in which a liquid changes into a solid on cooling. When water loses heat, it turns into ice.

- Melting → solid to liquid
- Evaporation → liquid to gas
- Condensation → gas to liquid

36. **Correct Answer: A) Solid to liquid**

Explanation:

Melting occurs when a solid gains heat and changes into a liquid.

Example: Ice melts to form water.

37. **Correct Answer: C) Gain or loss of heat**

Explanation:

Matter changes its state when it absorbs heat (melting, evaporation) or loses heat (freezing, condensation).

38. **Correct Answer: A) Solid**

Explanation:

Solids have closely packed particles. They have a definite shape and definite volume.

39. **Correct Answer: C) Gas**

Explanation:

Gas particles move freely and spread out to fill the entire container. They have neither fixed shape nor fixed volume.

40. **Correct Answer: B) Fixed volume but no fixed shape**

Explanation:

Liquids have a definite volume, but they take the shape of the container in which they are placed.

Integer type:

41. **Correct Answer: 7°C**

Explanation:

To convert Kelvin (K) to Celsius (°C), use the formula:

$$°C = K - 273$$

So,

280 – 273 = 7°C

42. **Correct Answer: Three**

Explanation:

Matter mainly exists in three states:

- Solid
- Liquid
- Gas

43. **Correct Answer: Two**

Explanation:

Mixtures are divided into two types:

1. Homogeneous mixture (uniform composition)
2. Heterogeneous mixture (non-uniform composition)

Previous year questions:

1. **Correct Answer:D**

   Explanation:

   Melting Point (i): This is the first horizontal plateau where the solid turns into a liquid. Looking at the y-axis, this occurs at 15°C.

   Boiling Point (ii): This is the second horizontal plateau where the liquid turns into a gas (vapour). Looking at the y-axis, this occurs at 110°C.

   Point 'c' (iii): This point is on the upward slope *between* the melting and boiling points. In this region, the wax has completely melted and is in its liquid state, increasing in temperature.

   Point 'e' (iv): This point is on the upward slope *after* the boiling plateau. Here, the wax has completely turned into a gas (or vapour) and is continuing to heat up.

2. **Correct Answer: Statements I and II are incorrect. Therefore, the correct option is C.**

   Explanation:

   i. Change 1 requires a decrease in temperature and an increase in pressure...: Incorrect. Sublimation requires an *increase* in temperature and usually a *decrease* in pressure.

   ii. Change 3 can be carried out by decreasing pressure and increasing temperature: Incorrect. Condensation (Gas to Liquid) requires *increasing* pressure and *decreasing* temperature.

   iii. Change 2 can be brought by heating the given substance: Correct. Melting requires heat.

   iv. Change 4 can be carried out in the presence of heat: Correct. Evaporation requires heat.

3. **Correct Answer : B) Water.**

   Explanation:

   Water is the most common substance on Earth that naturally exists in all three physical states within a normal range of environmental temperatures and pressures.

   The Three States of Water

   - Solid: Known as ice, found in glaciers, snow, and polar ice caps.
   - Liquid: Known simply as water, found in oceans, rivers, and rain.
   - Gas: Known as water vapor or steam, present in the atmosphere as humidity.

4. **Correct Answer: (B) Carbon dioxide.**

   Explanation

   In the process of respiration, gaseous exchange occurs in the alveoli of the lungs. Carbon dioxide (CO_2) diffuses from the blood (where its concentration is higher) into the lungs to be exhaled. Simultaneously, Oxygen (O_2) diffuses from the lungs into the blood to be transported to body tissues.

5. **Correct Answer: B) a,b,c.**

   Explanation

   Rising temperature increases the kinetic energy of particles, leading to:

   - Evaporation of liquid: More molecules gain enough energy to escape the surface.
   - Sublimation of solid: Increased energy allows solids to transition directly into gas.
   - Solubility of solute in water: For most solids, solubility increases as temperature rises because it helps break down the solute's lattice structure.

3. Sorting of Materials into Groups

Solutions

Worksheet – 1

1. **Correct answer – A**

   Transparent materials allow light to pass through them, so objects can be seen clearly.

2. **Correct answer – B**

   Hard materials resist cutting, bending, or scratching.

3. **Correct answer –A**

   A translucent object lets some light pass through, but we cannot see clearly through it.

4. **Correct answer – C**

   Ice floats on water, so it is lighter (less dense) than water.

5. **Correct answer –C**

   Chalk powder is insoluble in water. It does not mix completely with water.

6. **Correct answer –B**

   Matter is anything that has mass and occupies space (volume).

7. **Correct answer – D**

   Diamond is the hardest naturally occurring substance.

8. **Correct answer –B**

   Materials without shine are called dull or without lustre.

9. **Correct answer –A**

   Oiled paper lets some light pass through, so it is translucent, not opaque.

10. **Correct answer – B**

    In SI units, kilogram is written as kg with a space between the number and the unit.

Worksheet – 2

1. **Correct answer –B**

   It means arranging objects into groups based on similar properties.

2. **Correct answer – D**

   Sand does not dissolve in water, while sugar, salt, and copper sulphate dissolve.

3. **Correct answer – C**

   Boat is not a flower, but the others are flowers.

4. **Correct answer –C**

   The substance from which an object is made is called a material.

5. **Correct answer –B**

   Grouping helps us arrange things neatly and find them easily.

6. **Correct answer – C**

   A steel spoon is made of steel, while the others are made of different materials.

7. **Correct answer – A**

   A pen is usually made of two main materials, such as plastic and metal.

8. **Correct answer –B**

   An object is anything that we can see and touch.

9. **Correct answer –D**

   Humidity is about moisture in the air, not a property for grouping materials.

10. **Correct answer –A**

    Wood is lighter than water, so it floats. Soil, chalk powder, and glass sink.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer –A**
   Soft materials can be easily pressed, cut, bent, or scratched.
2. **Correct answer – A**
   Materials that look shiny are said to have lustre.
3. **Correct answer – C**
   Chair, bed, table, and cupboard are furniture, while a baby is not.
4. **Correct answer –C**
   A gas cylinder is much heavier than a feather, eraser, or leaf.
5. **Correct answer – A**
   Opaque materials do not allow light to pass through them, so we cannot see through them at all.
6. **Correct answer –A**
   A sugar cube is not hard, dissolves in water, and is not transparent. A glass sheet is hard, does not dissolve in water, and is transparent. An eraser is soft, insoluble in water, and opaque.
7. **Correct answer –B**
   Mustard oil does not mix with water and floats. Steel is hard and shiny. Wood is opaque. Mercury is a metal that is liquid at room temperature.
8. **Correct answer –B**
   Density shows how closely the particles of a substance are packed together.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. **Correct answer – A, C, D**
   Air is transparent.
10. **Correct answer – A, B, D**
    Iron is hard.
11. **Correct answer –A, B**
    Salt and sugar dissolve in water.
12. **Correct answer – B and C**
    There is space between number and unit
13. **Correct answer – A, C, D**
    Bicycle is made of more than one material.

ASSERTION AND REASNING TYPE

14. **Correct answer – A**
    The assertion is true because we frequently group items (like kitchen utensils by material or books by size/subject) to stay organized. The reason is also true and explains why we perform the classification mentioned in the assertion: it transforms a chaotic collection of items into a structured system that is much easier for us to navigate and understand.
15. **Correct answer –C**
    Many objects can be made from different materials (for example, chairs can be wood, plastic, or metal). But objects are often made using more than one material, so the reason is incorrect.
16. **Correct answer –B**
    Kerosene and water form separate layers because they are immiscible. The reason talks about heavy materials sinking, which is true, but it does not explain why they do not mix.
17. **Correct answer –C**
    Translucent materials let some light pass, so we cannot see clearly. Transparent materials allow almost all light to pass, so they are not the same.
18. **Correct answer –D**
    Sand is insoluble in water, so it does not dissolve. Therefore, the assertion is wrong, while the reason is correct.

MATRIX MATCH TYPE

19. **Correct answer –a-Q, b-S, c-P, d-R**
    Glass lets light pass clearly, so it is transparent. Oiled paper allows some light, so it is translucent. Wood blocks light, so it is opaque. Salt dissolves in water.

20. **Correct answer – a-Q, b-S, c-P, d-R.**
Litre is the unit of volume. Kilogram is the unit of mass. Metre is the unit of length. Degree Celsius is the unit of temperature.

COMPREHENSHION TYPE

Passage – I

21. **Correct answer –B.**
Solubility tells us whether a substance can dissolve in a liquid like water.
22. **Correct answer –C.**
Salt is soluble in water, while sand, oil, and stone do not dissolve.
23. **Correct answer – C.**
Sand is insoluble, so it remains undissolved and settles at the bottom.

Passage – II

24. **Correct answer –B.**
When items are grouped, they are arranged neatly and are easier to manage.
25. **Correct answer –C.**
Keeping similar things together helps us see differences and similarities easily.
26. **Correct answer –B.**
Paintbrushes and colours are materials commonly used for drawing or painting.

Previous Year Questions

1. **Correct answer –B.**
Magnifying glass and fresh air let light pass clearly. They are transparent.
Butter paper lets some light pass. It is translucent.
Rubber ball, newspaper, and leather bag block light. They are opaque.
2. **Correct answer –A.**
Polythene can be shaped easily. Gold is heavy so it sinks. Sugar dissolves.
Feather is light so it floats. Frosted glass lets some light pass.
3. **Correct answer –D.**
Gold ring is opaque, sinks, and shiny. That means P is gold
Wooden block is opaque, does not sink, not shiny. That means Q is wooden block.
Brick is opaque, sinks, not shiny. That means R is brick.
4. **Correct answer –C.**
Only sugar and honey dissolve completely. Sand and chalk powder remain undissolved.
5. **Correct answer –C.**
Soft substances are easily scratched or pressed.
6. **Correct answer –C.**
A dry leaf is light, so it floats instead of sinking.
7. **Correct answer –B**
A magnifying glass is transparent, not translucent. Muslin cloth lets some light through, so it is translucent. A mirror reflects light and you cannot see through it, so it is opaque.
8. **Correct answer – D.**
Copper coin is hard, opaque, sinks, insoluble. Clear glass is hard, transparent, sinks, insoluble. Cotton ball is soft, opaque, initially floats then sinks, insoluble.
9. **Correct answer –C.**
Vaseline is an oil-based substance that does not mix with water, while a metal spoon provides the necessary hardness and metallic luster. Cardboard is a classic opaque material that blocks all light, satisfying the third requirement. Finally, mercury is the only material listed that is both a liquid at room temperature and significantly denser (heavier) than water.
10. **Correct answer –B.**
The question asks where students can see the mark "clearly or partially." While P is clear, option B correctly identifies that Slide Q allows for partial visibility because the paint has made it translucent.

4. Separation of Substances and Mixtures

Solutions

Worksheet – 1

1. **Correct answer – B) Mixture**
   Explanation: Air contains nitrogen, oxygen, carbon dioxide, water vapour, etc., mixed together without chemical combination.
2. **Correct answer – B) Heterogeneous mixture**
   Explanation: In a heterogeneous mixture, the components are unevenly distributed and can often be seen separately.
3. **Correct answer – B) Homogeneous mixture**
   Explanation: Brass is an alloy of copper and zinc, uniformly mixed throughout.
4. **Correct answer – C) Gun powder**
   Explanation: Gun powder is a mixture of sulphur, charcoal, and potassium nitrate.
5. **Correct answer – C) Heterogeneous mixture**
   Explanation: Milk is a colloid (fat dispersed in water), so it is considered a heterogeneous mixture.
6. **Correct answer – B) Any ratio**
   Explanation: In mixtures, components can be mixed in any proportion.
7. **Correct answer – B) Mixture**
   Explanation: Mixtures are formed without chemical reaction and in variable proportions.
8. **Correct answer – A) Homogeneous**
   Explanation: A pure substance has uniform composition and properties throughout.
9. **Correct answer – A) F**
   Explanation: Fluorine (F) is a gaseous element, while calcium, iron, and cobalt are solids.
10. **Correct answer – B) Heterogeneous solid–solid mixture**
    Explanation: Charcoal and salt remain separate and are not uniformly mixed.

Worksheet-2

1. **Correct answer – C) Threshing**
   Explanation: Threshing is the process of separating grains from stalks by beating or hitting.
2. **Correct answer – A) Winnowing**
   Explanation: Winnowing uses wind to separate lighter husk from heavier grains.
3. **Correct answer – B) Handpicking**
   Explanation: In handpicking, unwanted substances like stones or spoiled grains are removed manually by hand.
4. **Correct answer – D) Sieving**
   Explanation: Sieving separates particles based on size using a sieve. Smaller particles pass through, while larger ones remain on the sieve.
5. **Correct answer – D) Water containing sand**
   Explanation: Filtration separates an insoluble solid from a liquid. Sand is insoluble in water, so it can be separated using filter paper.
6. **Correct answer – C) To note the boiling point**
   Explanation: The thermometer helps monitor the temperature at which the liquid boils.
7. **Correct answer – B) Purify liquids**
   Explanation: Distillation removes impurities from liquids based on boiling points.
8. **Correct answer – C) Evaporation**
   Explanation: When salt water is heated, water evaporates and solid salt is left behind
9. **Correct answer – C) Magnetic and non-magnetic substances**
   Explanation: Magnetic substances like iron are separated using a magnet.

10. **Correct answer – B) Distillation**
Explanation: A Liebig condenser cools vapours during distillation and converts them back into liquid.

Worksheet-3

1. **Correct answer – B) Insoluble solid material which settles down in the sedimentation process**
Explanation: Sediment is the heavy, insoluble solid that settles at the bottom of a liquid when left undisturbed.
2. **Correct answer – B) Solubility and absorption**
Explanation: In chromatography, different substances move at different speeds because of their different solubilities in the solvent and their absorption on the stationary phase.
3. **Correct answer – C) It gives pure crystals of the substance**
Explanation: Crystallisation removes both soluble and insoluble impurities and forms pure, well-defined crystals.
4. **Correct answer – C) Centrifugation**
Explanation: Centrifugation separates substances based on density by spinning them at high speed, commonly used in laboratories.
5. **Correct answer – C) Sublimation**
Explanation: Sublimation is the process in which a solid changes directly into vapour on heating and then back to solid on cooling.
6. **Correct answer – C) Colloid**
Explanation: Loading (coagulation) happens in colloids when particles lose charge and settle.
7. **Correct answer – B) Centrifugation**
Explanation: Centrifugation spins the curd rapidly, causing lighter cream to separate from heavier liquid.
8. **Correct answer – B) Salt**
Explanation: Salt can be purified by crystallisation as it forms pure crystals on cooling a concentrated solution.
9. **Correct answer – B) Components of a mixture dissolved in a solvent**
Explanation: Chromatography separates dissolved substances like dyes or pigments based on their movement with a solvent.
10. **Correct answer – D) Sedimentation**
Explanation: Sedimentation allows heavy insoluble particles to settle at the bottom.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C) Hand picking**
Explanation: Spoiled vegetables are easily visible and can be removed manually by hand.
2. **Correct answer – B) Sieving**
Explanation: Sieving separates materials based on size using a sieve with holes of fixed size.
3. **Correct answer – D) Evaporation**
Explanation: Seawater is allowed to evaporate, leaving behind solid salt.
4. **Correct answer – B) Evaporation**
Explanation: Evaporation occurs at temperatures below boiling point and happens slowly.
5. **Correct answer – D) Distillation**
Explanation: Distillation removes salts from seawater by evaporation and condensation.
6. **Correct answer – B) One of the components is solid and is insoluble in the liquid**
Explanation: Filtration separates an insoluble solid from a liquid using filter paper.
7. **Correct answer – D) Filtration**
Explanation: Silver chloride is insoluble in water, so it can be removed by filtration.

8. **Correct answer – C) The insoluble solid is completely removed, which is not possible in case of decantation**
   Explanation: Filtration removes even fine insoluble particles that may remain after decantation.
9. **Correct answer – B) Cornflakes in milk**
   Explanation: Cornflakes are insoluble and large enough to be separated using a strainer or filter.
10. **Correct answer – C) Filtration**
    Explanation: Tea leaves are insoluble in milk and can be removed using a strainer.
11. **Correct answer – B) Use a magnet to attract the iron filings**
    Explanation: Iron filings are magnetic, while sand is non-magnetic.
12. **Correct answer – B) Let the water evaporate**
    Explanation: Water evaporates on heating, leaving salt behind.
13. **Correct answer – B) Mobile phase**
    Explanation: The mobile phase carries the substances along the stationary phase.
14. **Correct answer – B) Obtain pure solid crystals from a solution**
    Explanation: Crystallisation forms pure crystals by cooling a concentrated solution.
15. **Correct answer – C) Mining industry**
    Explanation: Magnetic separation is used to separate magnetic ores from non-magnetic impurities in mining.
16. **Answer: B) Alum**
    Explanation: Alum neutralises the charge on colloidal impurities, causing them to coagulate and settle.
17. **Correct answer – D) Magnesium**
    Explanation: Magnesium is an element, while duralumin, magnalium, and bell metal are alloys (mixtures).
18. **Correct answer – A) Phosphorous**
    Explanation: Phosphorus is used on the striking surface of matchsticks.
19. **Correct answer – C) Gold, Mercury**
    Explanation: An amalgam is a mixture of mercury with another metal.
20. **Correct answer – A) Selenium**
    Explanation: Selenium shows properties of both metals and non-metals, so it is classified as a metalloid.
21. **Correct answer – B) A compound**
    Explanation:
    - Electrolysis is a method used to break down compounds into simpler substances (usually elements) using electricity.
    - The fact that the sample conducts electricity means it is an ionic compound or a compound in molten/dissolved state.
    - Mixtures and alloys cannot be broken into elements by electrolysis, and elements cannot be broken down further.
22. **Correct answer – C) Distillation — boiling point difference**

    Explanation:
    - Distillation: Separates liquids based on differences in boiling points.
    - Example: Water and alcohol mixture.
    - Chromatography: Separates components based on differences in solubility and adsorption, not density or boiling point.
    - Other options are incorrect because they mismatch the principle with the technique.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

1. **Correct answer – A) One of the components is soluble in a specific solvent.**

   Explanation: If one solid dissolves in a particular solvent and the other does not, the soluble solid can be separated by dissolving, filtering, and then evaporating the solvent.

2. **Correct answer – A,B,C,D**
   Explanation:
   - Magnetic separation (A) separates magnetic solids (like iron) from non-magnetic ones.
   - Gravity method (B) separates solids based on their different densities (e.g., separating sand from sawdust in water).
   - Fractional crystallization (C) separates mixtures of soluble solids that have different solubilities in the same solvent.
   - Sublimation (D) separates a solid that turns directly into gas (like iodine or camphor) from one that does not.
3. **Correct answer – A) The insoluble solid material which settles down when a suspension is allowed to stand undisturbed is called sediment.**
   Explanation: In sedimentation, heavier insoluble particles settle at the bottom and are called sediment.
4. **Correct answer – A, B and C (Water, Alcohol, A solvent mixture)**
   Explanation: The mobile phase is the solvent that moves through the stationary phase. Water, alcohol, or a mixture of solvents can act as the mobile phase. Filter paper is the stationary phase, not the mobile phase.
5. **Correct answer – A, B and C (Ink, Dyes, Colored pigments in leaves)**
   Explanation: Chromatography is used to separate colored substances dissolved in a solvent. Sand and iron filings are separated by magnetic separation, not chromatography.
6. **Correct answer – A, B, C**
   Explanation: Homogeneous mixtures have uniform composition throughout.
7. **Correct answer – A, B, D**
   Explanation: Heterogeneous mixtures have visible and unevenly distributed components.

ASSERTION AND REASNING TYPE

8. **Correct answer – A) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**

   Explanation: Filtration works because sand does not dissolve in water and its particles cannot pass through the tiny pores of the filter paper.
9. **Correct answer – A) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**

   Explanation: Handpicking is used when the unwanted material is visible and can be easily separated by hand.
10. **Correct answer – A) Both Assertion (A) and Reason (R) are true and R is the correct explanation of A**

    Explanation: Mud particles are heavier and insoluble, so they settle at the bottom (sedimentation), after which clear water is poured off (decantation).
11. **Correct answer – C) A is true but R is false**

    Explanation: because impurities *raise* the boiling point rather than lower it.

MATRIX MATCH TYPE

12. **Correct answer – (I): A–4, B–3, C–2, D–1**

    Metal – Shiny, malleable, ductile, good conductor of heat & electricity.

    Example: Iron → 4

    Non-Metal – Dull, brittle, poor conductor.

    Example: Carbon → 3

    Metalloid – Has properties of both metals and non-metals.

    Example: Arsenic → 2

    Radioactive element – Elements that emit radiation naturally.

    Example: Polonium → 1

13. **Correct answer – (II): A–2, B–3, C–1, D–4**

A. Magnetic separation → 2. Iron attracted by magnet
Explanation: Magnetic separation works because magnetic substances like iron are attracted by magnets.

B. Sedimentation → 3. Heavier insoluble particles settle
Explanation: In sedimentation, heavier and insoluble particles settle at the bottom of a liquid.

C. Distillation → 1. Difference in boiling points
Explanation: Distillation separates substances based on the difference in their boiling points.

D. Sieving → 4. Separation based on particle size
Explanation: Sieving separates substances by size using a sieve

COMPREHENSHION TYPE

14. **Correct answer – C) Weight**

Explanation: Winnowing works because lighter particles are blown away by air, while heavier particles fall down.

15. **Correct answer – A) Husk and rice**

Explanation: Husk is lighter than rice grains, so it can be separated using winnowing.

16. **Correct answer – A) Lighter particles are blown away by air**

Explanation: Air carries away lighter particles, leaving heavier ones behind.

17. **Correct answer – B) Boiling point**

Explanation: Distillation separates liquids because they boil at different temperatures.

18. **Correct answer – C) Distillate**

Explanation: The vapour condenses back into liquid, which is called the distillate.

19. **Correct answer – A) Alcohol and water**
Explanation: Alcohol and water have different boiling points, so they can be separated by distillation.
Integer type:
Explanation:
Insoluble solids can be separated from liquids by:
1) Sedimentation and decantation
2) Filtration
So, the total number of methods = 2.

20. **Sodium chloride (NaCl) is a compound made of 2 different elements: sodium (Na) and chlorine (Cl).**
A compound is a substance formed when two or more elements chemically combine in a fixed ratio.
In NaCl, sodium (a metal) reacts with chlorine (a non-metal) to form a white crystalline solid.
Each formula unit of NaCl contains one atom of sodium and one atom of chlorine**,** so the compound is made from two elements**.**
Explanation:
Total mixture = 1 kg
Husk removed = 0.2 kg
Grains left = 1 − 0.2 = 0.8 kg

21. **The boiling point of pure water is 100°C at 1 atmospheric pressure (1 atm or 101.3 kPa).**
Explanation:
The boiling point is the temperature at which a liquid changes into a gas (vapour) throughout the liquid, not just at the surface.
For water at normal atmospheric pressure, this occurs at 100°C.
At this temperature, the vapor pressure of water equals the external atmospheric pressure**,** allowing bubbles of vapour to form inside the liquid.
If the pressure changes (like at high altitudes), the boiling point of water will also change. For example, at higher altitudes, water boils at less than 100°C**.**

Previous Year Questions

1. Correct answer –

D) II – Gas, X – Evaporation

Explanation (in one line):

Diagram I shows liquid water changing to diagram II (gas) by evaporation, and the reverse change occurs by condensation.

2. Correct answer – A) Glycerine and coconut oil

Explanation:

Miscible liquids mix completely with water.

Example*:* Glycerine mixes completely with water.

Immiscible liquids do not mix with water and form separate layers.

Example*:* Coconut oil does not mix with water.

3. Substance Q (insoluble solid with small particles)
Best method: Filtration

Substance P (soluble solid in water)
Best method: Evaporation **(**water evaporates, solid remains)

Substance R (immiscible liquid in water)
Best method: Separating funnel

4. Correct answer – D

Explanation: I. Larger solid particles can be separated from smaller solid particles by filtration.

Incorrect. Filtration is used to separate an insoluble solid from a liquid. To separate solid particles of different sizes, we use sieving.

IV.Grains can be separated from husk by threshing.

Incorrect. Threshing is used to separate grains from the stalks. To separate grains from the husk, the correct method is winnowing.

5. Correct answer – B

P-(iv), Q-(i), R-(v), S-(ii), T-(iii)

Mixture (Column I)
Method (Column II)

P. Kidney beans and wheat flour

(iv) Sieving

They have different particle sizes; the flour passes through while beans stay.

Q. Mustard oil and water

(i) Separating funnel

These are two immiscible liquids; the oil floats on top.

R. Camphor and salt

(v) Sublimation

Camphor turns directly into gas when heated, leaving salt behind.

S. Rice flour and iron pins

(ii) Magnetic separation

Iron pins are attracted to magnets, while flour is not.

T. Fat from milk

(iii) Centrifugation

High-speed spinning separates the lighter fat particles.

6. Correct answer – C. Kerosene and water

C. Kerosene and water: Both are liquids. They are immiscible (do not mix), but because both pass through filter paper, filtration cannot separate them. Instead, a separating funnel is used to separate them based on their different densities.

7. Correct answer – B. Sand is left in the china dish

Explanation:Sand is left behind in the dish because it does not sublimate.

8. Correct answer – C

Z (Sand): Insoluble in water; can be separated by filtration.

X (Sugar): Soluble in water; forms a solution that passes through the filter.

Y (Water): The liquid solvent that evaporates, leaving the sugar behind.

9. Correct answer – D :P-(ii); Q-(iv); R-(i); S-(iii).

P. Sawdust and water

(ii) Filtration:Separating an insoluble solid from a liquid.

Q. Sand and gravel

(iv) Sieving Separating solids of different sizes.

R.Rice and husk

(i) Winnowing Separating heavier components from lighter ones using wind.

S. Mustard oil and water

(iii) Separating funnel Separating two immiscible liquids.

10. Correct answer – D. Magnetic separation, addition of water, filtration and evaporation

The correct sequence to isolate X:

1. Magnetic Separation: Removes Y (the only magnetic substance).
2. Addition of Water: Dissolves X, while Z remains a solid.
3. Filtration: Removes the insoluble Z. The liquid (filtrate) now contains only water and dissolved X.
4. Evaporation: Removes the water, leaving behind pure solid X.

5. Water

Solutions

Worksheet – 1

1. **Correct answer – D) Solid, liquid, and gas**
   Explanation:
   Water is unique because it exists naturally as a solid (ice/snow), liquid (rain/oceans), and gas (water vapor).
2. **Correct answer – B) 2:1**
   Explanation:
   The chemical formula for water is H_2O, meaning there are two atoms of Hydrogen for every one atom of Oxygen.
3. **Correct answer – B) Does not conduct electricity**
   Explanation:
   Pure water lacks dissolved salts or ions. Without these "charge carriers," electricity cannot flow through it.
4. **Correct answer – B) It can dissolve many substances**
   Explanation:
   Due to its chemical polarity, water can dissolve more substances than any other liquid on Earth.
5. **Correct answer – D) All of the above**
   Explanation:
   Hygiene, sanitation, and laundry are all heavily dependent on a steady supply of water.
6. **Correct answer – B) Measure water for all daily activities**
   Explanation:
   To find total usage, you must sum up drinking, cooking, cleaning, and flushing, not just one category.
7. **Correct answer – C) Growing food and making fabrics**
   Explanation:
   While drinking is vital, water is also the backbone of agriculture (crops) and industry (textile manufacturing).
8. **Correct answer – B) Lakes, rivers, or wells**
   Explanation:
   Most municipal systems draw from surface water (rivers/lakes) or groundwater (wells/aquifers).
9. **Correct answer – C) Lakes and rivers**
   Explanation:
   While ice caps contain most of the world's freshwater, they aren't easily "accessible." Rivers and lakes are the primary human sources.
10. **Correct answer – C) 0.01%**
    Explanation:
    About 97% of Earth's water is salty. Most freshwater is locked in glaciers. Only a tiny fraction is liquid and accessible for us.

Worksheet-2

1. **Correct answer – A) Ice (or Snow)**
   Explanation:
   Material P is Water. Its solid state is ice or snow, its liquid state is water, and its gas state is water vapor.
2. **Correct answer – A & B**
   Explanation:
   Rainwater is usually harvested from rooftops or from open ground surfaces (like parks or driveways) to be stored or recharged.
3. **Correct answer – B) Rainwater harvesting**
   Explanation:
   This is the specific term for the collection and storage of rain, rather than allowing it to run off as waste.
4. **Correct answer – B) Increases**
   Explanation:
   By directing rainwater into the ground or storage, we increase the groundwater table, helping prevent dry wells.

5. **Correct answer – B) Outlet pipe**
   Explanation:
   In a filtration system, the "inlet" brings the dirty water in, and the outlet pipe carries the filtered water to the tank or ground.
6. **Correct answer – B) Small**
   Explanation:
   While useful, a single rooftop collects a small amount compared to the massive volumes found in rivers or large drainage basins.
7. **Correct answer – B) Vapour**
   Explanation:
   Liquid water turns into water vapour (gas) through the process of evaporation, even at room temperature.
8. **Correct answer – A & B**
   Explanation:
   Condensation is when gas turns to liquid. This causes droplets on cold glasses and the tiny water droplets that form fog.
9. **Correct answer – A) Water Cycle**
   Explanation:
   The Water Cycle is the movement of water through the environment using evaporation, condensation, and precipitation.
10. **Correct answer – D) Floods**
    Explanation:
    Using the logic from Q1: P is water, Q is rain. Lack of rain (R) causes drought; an excess of rain (S) causes floods.

Worksheet-3

1. **Correct answer – B) Increasing dissolved oxygen and removing gases like CO_2 and H_2S**
   Explanation:
   Aeration exposes water to air, which increases dissolved oxygen and helps remove unpleasant gases such as carbon dioxide (CO_2) and hydrogen sulphide (H_2S) that cause bad smell and taste.
2. **Correct answer – C) Make tiny suspended particles stick together to form flocs**
   Explanation:
   During coagulation, chemicals like alum are added to water. These cause very fine suspended particles to clump together into larger particles called flocs, which can be removed easily.
3. **Correct answer – B) Heavy flocs formed during coagulation**
   Explanation:
   In sedimentation, heavy flocs settle down at the bottom of the tank due to gravity, making the water clearer.
4. **Correct answer – B) Remaining small particles, bacteria, and parasites**
   Explanation:
   Filtration passes water through sand, gravel, or other filters to remove small suspended particles and many microorganisms that were not removed during sedimentation.
5. **Correct answer –B) Chlorine**
   Explanation:
   Chlorine is commonly used in the final stage because it effectively kills harmful microorganisms and continues to protect water during storage and distribution.
6. **Correct answer – C) Distillation**
   Explanation:
   Distillation involves boiling water and condensing the steam. This process removes dissolved salts, minerals, bacteria, viruses, and organic impurities.
7. **Correct answer – B) Covered, narrow-necked containers**
   Explanation:
   Covered, narrow-necked containers prevent contamination from dust, insects, and hands, keeping treated water safe for drinking.
8. **Correct answer – A) Aerating by pouring water between containers**
   Explanation:
   Boiling removes dissolved air, making water taste flat. Pouring water between containers reintroduces air, improving taste.
9. **Correct answer – B) Screening**
   Explanation:
   Screening is the first step where large objects like leaves, twigs, fish, and plastic are removed from raw water.

10. **Correct answer – B) To remove large debris like twigs and fish**
Explanation:
Screening protects equipment and further treatment processes by removing large floating and suspended materials from raw water.

Worksheet-4

1. **Correct answer –B) The substance that gets dissolved in a solvent**
Explanation:
A solute is the substance that dissolves in a solvent to form a solution, for example, salt in saltwater.
2. **Correct answer – B) Saltwater**
Explanation:
Saltwater is a liquid solution because salt (solid) is dissolved in water (liquid).
3. **Correct answer – B) Industrial waste**
Explanation:
Industries release untreated chemicals, acids, and heavy metals into water bodies, causing serious water pollution.
4. **Correct answer – B) Dehydration and diseases**
Explanation:
Insufficient clean water leads to dehydration and water-borne diseases like diarrhoea and cholera.
5. **Correct answer – A) Supports aquatic life and ecosystems**
Explanation:
Water provides habitats for aquatic plants and animals and supports entire ecosystems.
6. **Correct answer – B) Because it dissolves more substances than any other liquid**
Explanation:
Due to its polar nature, water can dissolve many ionic and polar substances, earning it the name universal solvent.
7. **Correct answer – D) Solids, liquids, and gases**
Explanation:
A solute can be in any state—solid (salt), liquid (alcohol), or gas (carbon dioxide).
8. **Correct answer – C) Water**
Explanation:
In salt solution, water is present in a larger amount and dissolves salt, so it is the solvent.
9. **Correct answer – C) Air**
Explanation:
Air is a mixture of gases like nitrogen, oxygen, and carbon dioxide, making it a gas solution.
10. **Correct answer – C) Saturated solution**
Explanation:
A saturated solution contains the maximum amount of solute that can dissolve at a particular temperature.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C) Rainwater**
Explanation:
Rainwater is the purest form of natural water because it is formed by natural distillation (evaporation and condensation). However, it can pick up dust and gases as it falls through the atmosphere.
2. **Correct answer – C) 1 and 3**
Explanation:
(1) Breath contains water vapor that hits the cool mirror and turns to liquid. (3) Water vapor in the air touches the cold glass and turns to droplets. Both are examples of condensation.
3. **Correct answer – D) Freezing**
Explanation:
Rainwater is liquid. Snow is solid (ice crystals). The transition from a liquid state to a solid state due to low temperatures is called freezing..
4. **Correct answer – B) Water evaporates only in the day...**
Explanation:
This statement is false. Evaporation happens at all times, even at night or in the shade, though it is much faster when the sun provides direct heat.

5. **Correct answer – D) Seawater**
   Explanation:
   Seawater contains the highest concentration of dissolved salts and minerals compared to lakes, ponds, or groundwater, making it the most "impure" naturally.
6. **Correct answer – A) 1**
   Explanation:
   Evaporation is faster when the temperature is higher. The handkerchief under the direct sun (1) receives the most heat energy, causing the water to turn into vapor fastest.
7. **Correct answer – B) 2**
   Explanation:
   Statement 2 is slightly inaccurate or "wrong" in context: Clouds are not formed *from* cooled air; they are formed from water vapor that condenses *because* the air cooled down.
8. **Correct answer – D) Transpiration**
   Explanation:
   Transpiration is specifically the biological process where plants release excess water into the atmosphere through tiny pores in their leaves called stomata.
9. **Correct answer – C) Water Cycle**
   Explanation:
   In the water cycle, only pure water evaporates from the salty ocean, leaving the salt behind. This vapor then condenses and falls as fresh rain.
10. **Correct answer – A) Evaporation**
    Explanation:
    Evaporation is Liquid ⟶ Gas.
    Condensation is Gas ⟶ Liquid. They are exact opposites in the phase-change process.
11. **Correct answer – B) Physical changes only**
    Explanation:
    Water changing from liquid to vapor and back to liquid is a physical change (state change). No new chemical substances are created; it remains H_2O.
12. **Correct answer – D) Condensation**
    Explanation:
    The humid air outside contains a lot of water vapor. When it hits the cold surface of your spectacles (cooled by the AC), the vapor turns back into tiny liquid droplets (mist).
13. **Correct answer – C) Rainwater harvesting**

    Explanation:

    This is the broad term for capturing and storing rainwater or helping it recharge the groundwater table via pits or tanks.
14. **Correct answer – A) Water**

    Explanation:

    Material P is Water because it is the only substance essential for all life that exists naturally in three states (Ice, Water, Vapor).
15. **Correct answer – B) Rain**

    Explanation:

    Form Q is Rain. Water falls from the clouds specifically in the form of liquid precipitation (rain) or frozen precipitation (snow/hail).
16. **Correct answer – C) Drought**

    Explanation:

    Lack of rain (Q) for a long period leads to a severe shortage of water, which is known as a Drought.
17. **Correct answer – B) Henry Cavendish**

    Explanation:

    In 1781, the British scientist Henry Cavendish proved that water was not an element but a compound made by combining hydrogen and oxygen.
18. **Correct answer – B) 1:8**

    Explanation:

    By mass, hydrogen has an atomic mass of 1 and oxygen has 16. In H_2O, the ratio is

    (2 X1)16, which simplifies to 1:8

19. Correct answer – B) 97.4%

Explanation:

Roughly 97.4% of all water on Earth is found in the oceans. This water is too salty for drinking, irrigation, or most industrial uses.

20. Correct answer – C) Larger surface area

Explanation:
Powdered sugar has a much larger surface area compared to sugar cubes. This allows more sugar particles to come in contact with water at the same time, so it dissolves faster.

21. Correct answer – C) Marine ecosystems

Explanation:
Oil spills spread over the surface of oceans and seas, blocking oxygen and sunlight. This harms marine plants and animals, disrupting entire marine ecosystems.

22. Correct answer – B) Ocean → Vapour → Cloud → Rain → Rivers → Ocean

Explanation:

- Water cycle in nature:

1. Evaporation: Water from oceans, rivers, and lakes turns into water vapour.
2. Condensation: Water vapour forms clouds.
3. Precipitation: Clouds release water as rain, snow, or hail.
4. Collection: Rainwater flows into rivers and oceans, completing the cycle

23. Correct answer – B) Transpiration

Explanation:

- Transpiration is the process by which plants release water vapour into the air.
- Fewer trees → less transpiration → less water vapour → reduced cloud formation → less rainfall.
- This shows the role of forests in maintaining local rainfall patterns.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

24. Correct answer – A, B, and C

Explanation:

Dams are multipurpose. They store water for irrigation (Agriculture), use the force of falling water to spin turbines for electricity (Hydropower), and provide a steady supply for drinking.

25. Correct answer – A, B, and D

Explanation:

Pure water is colourless, odourless, and tasteless. Statement C is incorrect because pure water is an insulator; it only conducts electricity if it contains dissolved salts (ions)

26. Answer: A, B, and C

Explanation:

Water droplets fall from clouds as Rain (liquid), Hail (frozen lumps), or Snow (ice crystals). Dew (D) is not a form of falling rain; it condenses directly on cool surfaces like grass.

27. Correct answer – A, B, and C
Explanation:
Dams regulate water flow, drainage prevents accumulation, and Rainwater Harvesting stores water for future use. Deforestation (D) actually increases floods because trees help absorb water.

28. Correct answer – C
Explanation:
Known as infiltration or percolation, this is the process where rain soaks through the layers of soil and rocks to refill the underground aquifers

29. Correct answer – A) Aeration

Explanation:
Aeration exposes water to air, which removes dissolved gases like hydrogen sulphide (H_2S) and carbon dioxide (CO_2) that cause bad odor and unpleasant taste in water.

30. Correct answer – All of the above (A, B, C, and D).

Explanation

- Boiling: Bringing water to a rolling boil is the most effective household method to kill bacteria, viruses, and parasites.
- Distillation: This process involves boiling water and condensing the steam, effectively leaving bacteria and other contaminants behind in the original container.
- Chemical disinfection: Using chlorine, iodine, or bleach tablets effectively kills most pathogenic bacteria.
- Filtration: Modern household filters, particularly those with very small pore sizes (like ceramic or membrane filters) and activated carbon, can physically remove bacteria from water.

31. Correct answer – All of the above (A, B, C, and D).

Explanation

- Wastewater treatment: Prevents untreated sewage and contaminated water from entering natural water bodies.
- Agricultural management and organic farming: Reduces the runoff of harmful chemical fertilizers and pesticides into groundwater and rivers.
- Proper waste disposal: Prevents solid waste and toxic materials from being dumped directly into or near water sources.
- Industrial regulations: Forces industries to treat their effluent to remove toxic chemicals before discharging it, significantly reducing pollution at the source.

ASSERTION AND REASNING TYPE

32. Correct answer – (A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
Explanation:
Water changes into water vapour at the surface even at room temperature. This process is called evaporation, which makes wet clothes dry and spilled water disappear.

33. Correct answer – (D) Assertion is incorrect but Reason is correct
Explanation:
Evaporation can occur at any temperature, not only at 100°C. Heat from sunlight or warm air helps water molecules escape into the air.

34. Correct answer – (D) Assertion is incorrect but Reason is correct
Explanation:
Drought is caused by insufficient rainfall, not heavy rainfall. Long dry periods reduce water availability in soil and groundwater.

35. Correct answer – (A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
Explanation:
When warm, moist air rises and cools, it cannot hold all the water vapour. The vapour condenses into tiny droplets, forming clouds.

MATRIX MATCH TYPE

36. A-2, B-1, C-3, D-4
Explanation:

- Evaporation is the process by which water changes into vapour at temperatures below its boiling point.
- Transpiration is the loss of water vapour from the leaves of plants.
- Condensation occurs when water vapour cools and forms tiny droplets, leading to cloud formation.
- Precipitation is the falling of water from clouds in the form of rain, snow, or hail.

37. **A-2, B-1, C-3, D-4**
Explanation:
- Drought occurs when an area receives little or no rainfall for a long time.
- Floods happen due to heavy rainfall, causing water to overflow and submerge land.
- Rainwater harvesting involves collecting and storing rainwater to recharge groundwater.
- Groundwater overuse lowers the water table due to excessive withdrawal of underground water.
- COMPREHENSHION TYPE

COMPREHENSHION TYPE

I)

38. **Correct answer – B) Sunlight and warm air**
Explanation:
Heat from the Sun and warm air provides energy to water molecules, allowing them to change into water vapour. This process is called evaporation.

39. **Correct answer – B) Transpiration**
Explanation:
Transpiration is the process by which plants lose water in the form of water vapour through tiny pores (stomata) present on their leaves.

40. **Correct answer – C) Condensation of water vapour**
Explanation:
When water vapour cools, it changes into tiny water droplets. This process is called condensation, and these tiny droplets together form clouds.

II)

41. **Correct answer – A) Heavy rainfall**
Explanation:
Heavy or continuous rainfall causes rivers and water bodies to overflow, leading to flooding of nearby land areas.

42. **Correct answer –: A,B and C**
Explanation:
Soil moisture: This is often the first to be affected. Agricultural drought begins when soil moisture levels drop below the threshold needed for plants to survive.
Groundwater levels: As drought persists, the lack of rain prevents recharge (water soaking back into the ground). This causes the water table to drop, making wells go dry.
Crop growth: Low soil moisture and high temperatures directly stunt plant development, leading to wilting, reduced yields, and total crop failure.

43. **Correct answer –: A) Increase groundwater availability**
Explanation:
Rainwater harvesting involves collecting and storing rainwater so that it can percolate into the ground and recharge groundwater levels.
Integer type:
1. Answer: 97%
Explanation: Most of Earth's water is in the oceans and seas.
- Total water on Earth ≈ 100%
- Water in oceans and seas ≈ **97%**

44. **Correct answer –: less than 1 percent**
- **Explanation:** Only a small fraction of water on Earth is freshwater, and an even smaller fraction is easily accessible in lakes, rivers, and groundwater.
- Freshwater available for human use ≈ **2.5%-3%** of total water.

45. **Correct answer – 1 metre**
Explanation: In urban and semi-urban areas, recharge wells are used to collect rainwater and recharge groundwater. Typical recharge wells are about 0.5–1 metre in diameter, but most commonly: 1 metre.

46. **Correct answer – ≈68%**
- Explanation**:** The majority of Earth's freshwater is trapped in glaciers and ice caps.
- Percentage of freshwater frozen ≈ 68–70%

6. Air Around Us

Solutions

Worksheet – 1

1. **Correct Answer: C**
   The envelope of air surrounding the Earth is known as the atmosphere.
2. **Correct Answer: A**
   Air has no colour; it cannot be seen by our eyes.
3. **Correct Answer: C**
   Air has mass (weight) and pushes in every direction, which is called pressure.
4. **Correct Answer: A**
   Wind always moves from areas where pressure is higher to where it is lower.
5. **Correct Answer: A**
   Air from below pushes the paper upward and supports it.
6. **Correct Answer: B**
   The amount of gases, water vapour, and dust in air is not fixed, so it is a mixture.
7. **Correct Answer: C**
   Air can be liquefied and its gases separated without chemical change.
8. **Correct Answer: B**
   Oxygen supports burning. When it is used up, the flame stops.
9. **Correct Answer: C**
   Water vapour in the air cools and condenses into droplets.
10. **Correct Answer: B**
    Carbon dioxide reacts with lime water to form a white substance (calcium carbonate).

Worksheet – 2

1. **Correct Answer: B**
   About one-fifth of air is oxygen, which is roughly 20.9%.
2. **Correct Answer: D**
   The amount of water vapour changes with place, temperature, and weather.
3. **Correct Answer: C**
   Helium is an inert (noble) gas and does not react easily.
4. **Correct Answer: C**
   Plants use carbon dioxide, water, and sunlight to make food.
5. **Correct Answer: C**
   Kerosene is commonly used as a domestic fuel in such stoves.
6. **Correct Answer: C**
   The main component of CNG is methane.
7. **Correct Answer: D**
   Living beings breathe out carbon dioxide as a waste gas.
8. **Correct Answer: B**
   Oxygen helps living organisms release energy from food.
9. **Correct Answer: B**
   In photosynthesis, plants make food using sunlight.
10. **Correct Answer: B**
    These bacteria convert atmospheric nitrogen into a usable form for plants.

Worksheet – 3

1. **Correct Answer: B**
   Air pollution happens when harmful substances, called pollutants, mix with clean air.
2. **Correct Answer: C**
   Pollutants are unwanted harmful materials that make the air dirty and unsafe.
3. **Correct Answer: B**
   Cars, buses, and trucks release smoke and harmful gases from their exhaust pipes.
4. **Correct Answer: B**
   Rotting plants and animals give off methane, which adds to air pollution.

5. **Correct Answer: C**
Dust and smoke irritate our lungs and may lead to asthma or shortness of breath.
6. **Correct Answer: B**
Polluted gases react with water in clouds and fall as acid rain, which damages crops and buildings.
7. **Correct Answer: B**
Too much carbon dioxide, methane, and other gases trap heat, making the planet warmer.
8. **Correct Answer: C**
Ozone acts like sunglasses for Earth and blocks harmful UV rays from the Sun.
9. **Correct Answer: B**
Trees clean the air by taking in carbon dioxide. Fewer trees mean more pollution.
10. **Correct Answer: C**
Renewable energy like solar and wind produces little or no harmful smoke.

Competitive Corner

SINGLE OPTION CORRECT TYPE:

1. **Correct Answer: D**
White anhydrous copper sulphate absorbs moisture from air and becomes blue hydrated copper sulphate.
2. **Correct Answer: C**
Air has about 21% oxygen and 78% nitrogen, which is roughly 1:4.
3. **Correct Answer: A**
The atmosphere (especially the ozone layer) blocks harmful ultraviolet radiation.
4. **Correct Answer: B**
Weather describes short-term atmospheric conditions.
5. **Correct Answer: B**
Nitrogen is the major component of air by volume.
6. **Correct Answer: B**
Oxygen supports burning, it was named by Antoine Lavoisier, and living beings use it for respiration.
7. **Correct Answer: B**
Lower outside pressure lets the air inside expand.
8. **Correct Answer: B**
Older refrigerators used chlorofluorocarbons, which harm the ozone layer.
9. **Correct Answer: A**
CO_2 is the least, nitrogen the highest.
10. **Correct Answer: C**
Typical question based on plant cell wall component.) Cellulose forms the main structural material.
11. **Correct Answer: A**
More CO_2 enhances the greenhouse effect, trapping heat.

ONE OR MORE THAN ONE CORRECT OPTION TYPE:

12. **Correct Answer: D**
Air pushes on objects from every side—up, down, and sideways.
13. **Correct Answer: A, B, C**
Air is needed for life. Air is present everywhere on land. Air takes up space. We cannot see air, so D is wrong.
14. **Correct Answer: A, C, D**
Air has mass (weight) and occupies space. It is a mixture of gases. It exerts pressure in all directions.
15. **Correct Answer: A, B, D**
Oxygen is a component of air. Water vapour is also present. Air is not a single substance; it is a mixture. Its composition can vary from place to place.

ASSERTION AND REASON TYPE

16. **Correct Answer: A**
Nitrogen oxides formed in air react with oxygen and water vapour to produce nitric acid. This dilute nitric acid comes down with rain and reacts with soil carbonates to form nitrates, so the reason correctly explains the assertion.
17. **Correct Answer: B**
Both statements are true, but the reason does not fully explain why respiration is similar to burning (the similarity is due to oxidation of food).

18. **Correct Answer: C**
The assertion is true because trees absorb carbon dioxide and improve air quality. The reason is false since CNG and electric vehicles reduce harmful emissions.
19. **Correct Answer: D**
Plants cannot use free atmospheric nitrogen directly. However, lightning helps convert nitrogen into compounds such as nitric oxide, which later form usable substances.
20. **Correct Answer: A**
In a mixture, substances are not in a fixed ratio and each keeps its own properties. Air shows these features, so it is a mixture.

MATRIX MATCHING TYPE

21. **Correct Answer: a-T, b-P, c-Q, d-S**
Oxygen forms about one-fifth of air. Nitrogen is the most abundant gas in the atmosphere. CO_2 is present only in a very small quantity. Argon and other noble gases together make up around 1%.
22. **Correct Answer: a-S, b-R, c-P, d-Q**
Oxygen is used by living beings for respiration and it supports combustion. Nitrogen is fixed into the soil with the help of bacteria in leguminous plants. CO_2 reacts with lime water to form calcium carbonate, making it milky. Water vapour (moisture) changes white copper sulphate into blue.

COMPREHENSION TYPE

Passage – I

23. **Correct Answer: A**
Digestion converts complex food into simpler substances that the body can absorb and use.
24. **Correct Answer: D**
Food contains carbohydrates like starch and sugar, as well as fats.
25. **Correct Answer: C**
Our body uses oxygen for respiration, so exhaled air has less oxygen, not more.

Passage – II

26. **Correct Answer: C**
The major gases in the atmosphere are nitrogen (~78%) and oxygen (~21%).
27. **Correct Answer: B**
Oxygen is about one-fourth the amount of nitrogen in air.
28. **Correct Answer: A**
Anhydrous copper sulphate turns from white to blue when it absorbs moisture from air.

INTEGER TYPE

29. **Correct Answer: 3**
One molecule of CO_2 has 1 carbon atom and 2 oxygen atoms → total = 3.
30. **Correct Answer: 6**
The formula of glucose is $C_6H_{12}O_6$, so it contains 6 carbon atoms.
31. **Correct Answer: 1**
Air contains about 78% nitrogen and 21% oxygen, which is approximately 4 : 1.

Previous Year Questions

1. **Correct Answer: D**
Nitrogen is a gas, but it does not support burning. It is also not released during respiration, so it corresponds to S.
2. **Correct Answer: C**
Layer X (Mesosphere) is actually the coldest layer of the atmosphere, while Layer W (Thermosphere) is the hottest.
3. **Correct Answer: B**
Planting trees, using cleaner fuels like LPG/CNG, and checking vehicle emissions all help reduce air pollution. Treating sewage before releasing it into water mainly controls water pollution, not air pollution.
4. **Correct Answer: D**
From the graph, Z ≈ nitrogen (78%), X ≈ oxygen (21%), and Y ≈ carbon dioxide (very little). Plants use carbon dioxide for photosynthesis, not nitrogen, and oxygen is not a fertiliser component, so D is incorrect.

7. Changes Around US

Solutions

Worksheet – 1

1. **Correct Answer: D) The ring changes its shape and the change cannot be reversed.**
   Explanation:
   When an iron ring is heated, it expands and on cooling it comes back almost to its original size. This is a reversible change.
2. **Correct Answer: A) Hardening of cement**
   Explanation:
   Once cement hardens, it cannot be brought back to its original form.
   Freezing ice cream, opening a door, and melting chocolate are reversible changes.
3. **Correct Answer: D) Expands first, then on cooling contracts and fits onto the wooden wheel.**
   Explanation:
   On heating, the iron rim expands and fits easily.
   On cooling, it contracts tightly, holding the wooden wheel firmly.
4. **Correct Answer: B) Melting of ghee**
   Explanation:
   Melted ghee can be solidified again by cooling, so it is reversible.
   Cutting trees, burning a candle, and blooming of a flower are irreversible.
5. **Correct Answer: D) Mixing salt in water is a change that can be reversed by evaporation.**
   Explanation:
   When salt solution is heated,water evaporates and salt is left behind, so the change is reversible.
6. **Correct Answer: C) Can be reversed and cannot be reversed, respectively.**
   Explanation:
   Rolling dough → can be changed back into a ball (reversible).
   Baking chapati → cannot be changed back into dough (irreversible).
7. **Correct Answer: C**
   Explanation: Burning produces new substances and cannot be reversed.
8. **Correct Answer: C**
   Explanation: Torn paper cannot be restored to its original form.
9. **Correct Answer: B**
   Explanation: Both melting and boiling can be reversed by cooling.
10. **Correct Answer: D) Freezing of water to ice**
    Explanation:
    Ice can melt back into water, so the change is temporary and reversible.
    Other options are permanent changes.

Worksheet-2

1. **Correct Answer: B) Fast change**
   Explanation:
   Magnesium burns very quickly in air with a bright flame, so it is a fast change. It is also irreversible, but the question focuses on the speed.
2. **Correct Answer: B) Rusting of iron**
   Explanation:
   Rusting takes place slowly over a long period of time, often days or months.
3. **Correct Answer: B) Fast change**
   Explanation:
   The matchstick burns quickly once lit, so it is a fast change. It is also irreversible.
4. **Correct Answer: B) Slow change**
   Explanation:
   Aging happens gradually over time, so it is a slow change.
5. **Correct Answer: B) Fast changes**
   Explanation:
   Explosions occur suddenly and within a very short time, making them fast changes.

6. **Correct Answer: B) Gradual changes**
   Explanation:
   Slow chemical changes occur gradually, not instantly.
7. **Correct Answer: B) Burning of paper**
   Explanation:
   Burning of paper happens quickly and produces new substances like ash and gases.
8. **Correct Answer: A) Rusting of iron**
   Explanation:
   Rusting is a very slow chemical process that takes a long time.
9. **Correct Answer: A) Fast change**
   Explanation:
   Wood burns relatively quickly and the change is irreversible, so it is a fast change.
10. **Correct Answer: B) Slow change**
    Explanation:
    Patina forms slowly over a long period due to reaction with air and moisture.

Competitive corner

SINGLE CORRECT ANSWER TYPE

1. **Correct Answer: A) Slow**
   Explanation:
   A slow change happens over a long period of time. It does not occur immediately, for example rusting of iron or growth of a plant.
2. **Correct Answer: A) Reversible change**
   Explanation:
   In a reversible change, the original substance can be obtained again.
   Example: melting of ice into water and freezing it back into ice.
3. **Correct Answer: B) A puppy growing into a dog**
   Explanation:
   Growth of a puppy into a dog takes a long time, so it is a slow change.
   Other options like ripening and rotting also take time, but growth is the most appropriate example here.
4. **Correct Answer: A) Bursting of a cracker**
   Explanation:
   Bursting of a cracker happens in a very short time, almost instantly, so it is a fast change.
5. **Correct Answer: B) Fast**
   Explanation:
   Any change that happens quickly or instantly, such as burning or bursting, is called a fast change.
6. **Correct Answer: C) State, taste, and smell**
   Explanation: When milk turns into curd, it becomes thicker (change in state), sour (change in taste), and develops a different smell. Hence, multiple properties change.
7. **Correct Answer: B) Bacteria cannot grow in cold milk**
   Explanation: The bacteria (Lactobacillus) needed to convert milk into curd do not grow well at very low temperatures, so curd formation does not occur.
8. **Correct Answer: C) Winter**
   Explanation: In winter, the Sun appears for fewer hours in the sky, resulting in shorter days and longer nights.
9. **Correct Answer: C) May**
   Explanation: May is a summer month in the Northern Hemisphere, when days are longer and nights are shorter.
10. **Correct Answer: C) South of east**
    Explanation: During winter (December), the Sun rises slightly south of the east direction due to the tilt of the Earth's axis.
11. **Correct Answer: B) Both melting ice and sugar dissolving are physical changes**
    Explanation:

    - Melting ice → solid water changes to liquid water. No new substance is formed. Physical change.
    - Sugar dissolving → sugar molecules disperse in water but can be recovered by evaporation. No new substance is formed. Physical change.

12. **Correct Answer: B) Refrigeration is a physical change, slowing chemical changes**
Explanation:

- Refrigeration lowers temperature, which slows down chemical reactions like ripening and spoilage.
- The cooling itself is a physical change (temperature change, no new substances formed).
- Over time, fruits still undergo chemical changes (ripening, enzymatic reactions), just at a slower rate.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

13. **Correct Answer: A,C and D**

Explanation: Fast changes occur within seconds or minutes.

- Burning paper (A) and Combustion of kerosene (D) are rapid chemical reactions.
- An **Explosion (C)** is almost instantaneous

14. **Correct Answer: A) ,B and D**

- Explanation: Slow changes take hours, days, or years to complete.
- **Rusting (A)**, **Ripening (B)**, and the **Formation of patina (D)** (that green layer you see on old copper statues) all happen gradually over time.

15. **Correct Answer: B,C and D**

Explanation: An irreversible change is a "one-way street"—you cannot get the original materials back.

- **Burning wood (B)** turns it into ash and smoke.
- **Cooking an egg (C)** changes the protein structure permanently.
- **Rusting (D)** turns iron into iron oxide.

16. **Correct Answer: A,B and D**
Explanation: These are physical changes where the state of matter changes, but the substance stays the same.
    - **Melting (A)**, **Boiling (B)**, and **Freezing (D)** are all phase changes of water. You can move back and forth between ice, liquid, and steam just by changing the temperature.
17. **Correct Answer: A,B and D**
Explanation: These are all forms of Combustion or rapid oxidation.
    - **Kerosene (A)** and **Magnesium (B)** both produce intense heat and bright light when burned.
    - A **Firecracker (D)** releases energy as sound, light, and heat simultaneously

ASSERTION AND REASNING TYPE

18. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A.**

Explanation: Burning of paper is classified as a fast change because the reaction between the paper and oxygen occurs almost instantly once the ignition temperature is reached.

19. **Correct Answer: D) A is false, R is true.**

Explanation: The Assertion is false. Melting of ice is a reversible change because you can turn the resulting liquid water back into solid ice by freezing it.

20. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A.**

Explanation: Unlike an explosion or a flame, freezing water usually takes a significant amount of time (often hours in a domestic freezer), making it a slow change compared to instantaneous reactions.

21. **Correct Answer: B) Both A and R are true, but R is not the correct explanation of A.**

Explanation: The Assertion is true; using a knife to slice through fruit happens in a matter of seconds.

MATRIX MATCH TYPE

22. **Correct Answer:A → Q, B → P, C → R, D → S**
Q) Slow change Takes a long time to occur.
p) Fast change Happens almost instantly upon ignition.
R) Reversible change Can be changed back to ice by freezing
S) Irreversible change Chemical structure is permanently altered.
23. **Correct Answer:A → S, B → Q, C → R, D → P**
S) Irreversible change Kerosene turns into gases and energy; it cannot be recovered.
Q) Slow changeTakes several hours for bacteria to turn milk into curd.
R) Reversible changeVapor can be condensed back into liquid water.
p) Fast change. The chemical reaction happens instantly upon striking.

COMPREHENSHION TYPE

Passage-I

24. **Correct Answer: B) Irreversible change**
    - Explanation: When you cook an egg, the heat permanently alters the structure of the proteins (denaturation). Once an egg is boiled or fried, it is impossible to return it to its original liquid, raw state.
25. **Correct Answer: B) New substances are formed**
    - Explanation: Cooking is a chemical change. The heat causes the molecules in the egg to react and bond in new ways, creating a solid structure with different chemical properties than the raw egg.
26. **Correct Answer: B) Produce permanent new substances**
    - Explanation: By definition, an irreversible change results in the formation of one or more new substances that cannot be converted back into the original materials. Examples include burning wood into ash or rusting iron into iron oxide.

Passage-II

27. **Correct Answer: B) Fast change**
    - Explanation: When you strike a match, the chemical reaction (combustion) happens almost instantly, producing a flame in a split second.
28. **Correct Answer: B) Produce new substances rapidly**
    - Explanation: Fast changes occur within a very short duration, such as seconds or minutes. If the fast change is a chemical reaction (like combustion), it will produce new substances very quickly.
29. **Correct Answer: A) Burning paper**
    - Explanation: Burning paper takes only a few seconds to complete, making it a fast change.

Previous year questions:

1. **Correct Answer:B**
Explanation:
    - P: Melting of ice — This is a physical change (no new substance) and it is reversible (you can freeze it again).
    - Q: Shredding of paper — This is a physical change because it is still paper, but it is irreversible because you cannot realistically join the tiny fibers back into the original smooth sheet.
2. **Correct Answer:C**
Explanation: (i) Tearing of paper is a physical, irreversible change.
    - True (T): No new substance is formed, but you cannot restore the original sheet.

(ii) Burning of wood is a chemical, reversible change.
    - False (F): Burning is a chemical change, but it is irreversible; you cannot turn ash back into wood.

(iii) Melting of butter is a physical, reversible change.
    - True (T): It is a change of state (physical) and can be solidified again by cooling (reversible).

3. **Correct Answer:D**
   Explanation:
   - P. Rusting of iron → (i) Chemical, (iv) Irreversible: It forms iron oxide and cannot be undone.
   - Q. Expansion of metal rim → (ii) Physical, (iii) Reversible: Metal expands when heated and contracts when cooled.
   - R. Sharpening of pencil → (i) Chemical? No, this is likely (ii) Physical, (iv) Irreversible: Note: Option D matches R with (i) and (iii), which is a common error in some Correct Answer keys. However, looking at the logic of competitive exams like NSO, D is the intended Correct Answer because it correctly identifies the most distinct traits of P, Q, and S.
   - S. Cooking of food → (i) Chemical, (iv) Irreversible: Heat creates new chemical bonds that cannot be reversed.
4. **Correct Answer:A**
   Explanation:

Number	Process	Properties
1	Freezing of ice-crea	Desirable+Physical
2	Ripening of fruits	Desirable+Chemical
3	Burning of candle	Physical + Chemical

5. **Correct Answer:B**
   Explanation:
   B: Condensation (i) releases heat; Melting (ii) absorbs heat; Burning of coal (iii) is an irreversible chemical change; Dissolution of sugar (iv) is a reversible physical change.
6. **Correct Answer: B (II and IV only)**
   Explanation: II. Occurrence of day and night is a periodic and non-desirable change: Incorrect. While it is periodic (happens every 24 hours), it is a desirable change necessary for life on Earth
   IV. Breaking of glass is a physical and reversible change: Incorrect. While it is a physical change (it's still glass), it is irreversible because you cannot simply "un-break" the shards back into the original shape without melting and reforming it.

Process	Chemical?	Irreversible?	Slow?	Fits X?
I. Dissolving sugar	No (Physical)	No(Reversible)	Fast/ Med	No
II. Burning of paper	Yes	Yes	Fast	No
III. Growing of plant	Yes	Yes	Yes	Yes
IV. Digestion of food	Yes	Yes	Yes	Yes

7. **Correct Answer: B (I and II)**
   Explanation:

8. **Correct Answer: D (Iron and chemical, irreversible change)**
   Explanation:
   When a metal is dropped into blue copper sulphate ($CuSO_4$) and the solution turns green, a displacement reaction has occurred.
   - The Metal: When Iron (Fe) is added to copper sulphate, it displaces the copper to form Iron Sulphate ($FeSO_4$), which is green in color.
   - The Change: This is a Chemical change because a new substance is formed, and it is Irreversible under normal conditions because the iron has been chemically transformed.

   $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$
9. **Correct Answer: C**
   Explanation:
   - P (Not Physical + Not Desirable): Rusting of iron is a chemical change (not physical) and is undesirable.
   - Q (Physical + Irreversible): Shredding of paper changes the form (physical) but cannot be naturally undone to its original state (irreversible).
   - R (Physical + Reversible + Desirable): Glowing of a filament

bulb is physical (light/heat) and reversible by turning the switch off.

- S (Not Physical + Desirable): Curdling of milk is a chemical change (not physical) that is desirable when making yogurt or cheese.

10. **Correct Answer: B: (i) Undesirable, (ii) Non-periodic, (iii) Desirable, (iv) Slow.**
    Explanation:
    - (i)Rusting of iron and Spoilage of food: Both are Undesirable changes.
    - (ii) Rusting of iron and Making curd: Both are Irreversible changes (though curdling is desirable, rusting is not).
    - (iii) Making curd and Ripening of fruits: Both are Desirable changes and also Slow changes.
    - (iv) Spoilage of food and Ripening of fruits: Both are Slow changes.